

SYDNEY PENRITH, Australia

WMO: 947630

Lat: 33.7194S

Lon: 150.6783E

Elev: 25

StdP: 101.03

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	4.1	-2.3	3.1	17.0	-0.9	3.5	15.5	10.7	14.8	9.0	15.8	0.9	190	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	36.8	21.0	34.2	20.8	31.9	20.4	24.0	29.9	23.2	28.4	22.5	27.5	4.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	17.6	25.1	22.0	16.7	24.2	21.2	15.9	23.8	72.0	30.1	68.9	28.3	66.2	27.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.6	6.5	0.7	41.5	0.9	2.6	0.0	43.3	-0.6	44.8	-1.1	46.3	-1.8	48.2		
			0.4	25.7	0.9	1.1	-0.3	26.4	-0.8	27.1	-1.3	27.7	-1.9	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	24.4	23.5	21.9	18.7	15.0	12.4	11.6	12.8	16.0	18.7	20.8	23.1
	DBStd	5.14	3.16	2.98	2.46	2.35	2.14	1.94	1.89	2.04	2.62	3.07	3.40	3.14
	HDD10.0	10	0	0	0	0	0	3	6	2	0	0	0	0
	HDD18.3	818	0	1	2	23	106	177	209	172	80	34	13	2
	CDD10.0	3004	446	379	368	261	154	76	55	88	180	268	323	406
	CDD18.3	771	188	146	112	33	2	0	0	1	9	44	87	149
	CDH23.3	6216	1580	1065	661	195	14	0	2	8	163	478	778	1271
	CDH26.7	2605	757	443	208	42	1	0	0	1	44	185	341	585
Wind	WSAvg	2.5	2.9	2.8	2.5	2.2	1.9	2.1	2.1	2.4	2.6	2.7	2.9	2.8
Precipitation	PrecAvg	883	102	110	99	77	59	61	37	51	41	67	95	86
	PrecMax	1406	308	447	306	316	215	283	172	398	192	171	540	268
	PrecMin	447	5	11	5	6	0	1	0	0	0	0	5	2
	PrecStd	261	75	102	72	77	58	69	42	78	36	50	98	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	38.7	34.2	31.1	25.3	21.0	22.5	24.4	31.1	34.5	37.8	38.8
		MCWB	22.4	22.4	21.3	18.6	15.7	14.2	13.5	13.7	16.5	17.5	18.8	20.8
	2%	DB	36.8	34.5	31.0	27.4	23.0	19.1	19.5	22.0	27.6	31.3	33.6	35.2
		MCWB	21.9	22.3	21.2	18.1	15.7	13.1	12.0	13.1	15.4	17.3	19.5	20.5
	5%	DB	33.6	31.4	28.8	25.5	21.4	17.9	17.9	20.1	24.9	28.5	30.3	32.3
		MCWB	21.5	21.8	20.4	17.9	14.9	12.6	11.5	12.3	15.0	17.0	18.7	19.9
	10%	DB	30.9	29.1	26.9	23.7	20.0	16.8	16.5	18.4	22.6	25.6	27.7	29.8
		MCWB	21.2	20.8	19.8	17.3	14.2	12.2	11.0	11.5	14.2	16.3	18.3	19.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.0	25.5	24.2	21.2	18.6	16.3	16.0	16.0	18.3	20.7	23.0	24.0
		MCDB	31.6	32.4	28.0	25.4	21.2	18.2	18.2	20.4	24.0	26.4	28.7	30.4
	2%	WB	23.8	24.1	22.8	20.1	17.3	15.0	13.9	14.6	17.2	19.4	21.6	22.8
		MCDB	30.9	29.1	26.8	23.8	20.2	16.8	16.5	19.0	23.1	25.0	27.3	28.9
	5%	WB	23.1	23.1	21.9	19.2	16.2	14.3	12.7	13.5	16.2	18.5	20.7	22.0
		MCDB	28.7	27.6	25.8	22.7	19.4	16.1	15.5	17.8	21.7	24.3	26.0	27.7
	10%	WB	22.4	22.3	21.1	18.5	15.4	13.4	12.0	12.5	15.2	17.6	19.7	21.2
		MCDB	27.8	26.8	24.8	21.8	18.5	15.4	15.1	16.7	20.7	23.1	24.7	27.1
Mean Daily Temperature Range	5% DB	MDBR	11.3	9.9	9.6	10.4	10.9	9.9	11.3	12.3	12.9	12.4	11.3	11.6
		MCDBR	17.3	14.9	13.3	13.4	12.9	11.6	13.6	15.4	17.4	18.1	17.4	16.9
	5% WB	MCWBR	4.6	4.5	4.7	5.2	6.1	6.3	7.1	7.2	6.6	6.1	5.2	4.8
		MCDWR	13.1	11.6	11.1	10.9	10.6	8.6	10.7	12.8	14.1	13.4	13.2	13.3
Clear-Sky Solar Irradiance	taub		0.396	0.393	0.364	0.334	0.317	0.307	0.297	0.304	0.337	0.360	0.386	0.392
		taud	2.428	2.459	2.517	2.574	2.567	2.598	2.607	2.579	2.486	2.449	2.421	2.419
	Ebn at Noon		942	924	916	891	853	839	869	911	928	945	945	949
		Edh at Noon	123	115	101	85	76	69	71	83	102	114	123	125
All-Sky Solar Radiation	RadAvg		6.39	5.44	4.67	3.82	3.06	2.42	2.83	3.73	4.95	5.69	6.00	6.58
	RadStd		0.51	0.52	0.43	0.34	0.21	0.21	0.21	0.23	0.30	0.57	0.57	0.65

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	+0.44	N/A	N/A	+0.77	N/A	N/A	-2	-86	+185	+99	

Nomenclature: See separate page